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BACHELOR OF ENGINEERING (Hons) – AK3751

Mid-Semester Test

Semester 2, 2018

FLUIDS & THERMODYNAMICS
(Module – Thermodynamics A)

TIME ALLOWED: 1 hour and 15 minutes

MARKS: TOTAL: 36 marks

INSTRUCTIONS: Answer **ALL FOUR questions.**

All answers except diagrams and sketches ***must be in ink.***

Write your answers in this answer booklet.

Correcting fluid is not permitted.

Exam conditions apply!

NOTES: Marks allocated to each question are shown in brackets beside each question.

Candidates should show ***all working*** in calculations.

Useful Formulae can be found at the end of this exam paper.

Tables of Thermodynamic Data are supplied.

Question 1 – (3 marks)

Under what conditions (temperature, pressure, density) could the *ideal gas equation* be used to *accurately* predict properties for *steam*?

Question 2 – (9 marks)

Fill in the missing information in the following table for ***R134a***.

	T **(°C)**	***p*** **(kPa)**	***h*** **(kJ/kg)**	***x***	**Phase description**
a.			22.89		saturated liquid
b.		800	200		
c.	-20			0.3	
d.		1000	62.68		
e.	40	200			
f.	85		279.51		

Question 3 – (16 marks)

A piston-cylinder device (Figure 1) with a set of stops on the top contains 2.2 kg of saturated liquid water at 150 kPa. Heat is transferred to the water, causing some of the liquid to evaporate and move the piston upwards (at constant pressure). When the piston reaches the stops, the enclosed volume is 45 l. More heat is transferred until the pressure is doubled. Determine:

a) The amount of saturated liquid (if any), in kg, at the final state.

b) The final temperature.

c) The total work and heat transfer.

d) Show the process on a p–v diagram (with respect to saturation lines).

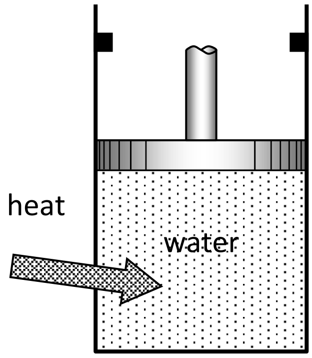


Figure 1

Question 3 (cont'd)

Question 3 (cont'd)

Question 3 (cont'd)

Question 4 – (8 marks)

An insulated rigid vessel with a volume of 3 m^3 is fitted with a resistance heater and a paddle wheel as shown in Figure 2. The vessel is initially filled with air at a temperature of 27°C and a pressure of 130 kPa. Now, a current of 4 A from a 230 V supply is passed through the resistance heater while the paddle wheel transfers energy at a rate of 150 J/s to the air. Determine the time taken for the air to reach a temperature of 227°C.

(State any assumptions made and also state any tables used for data).

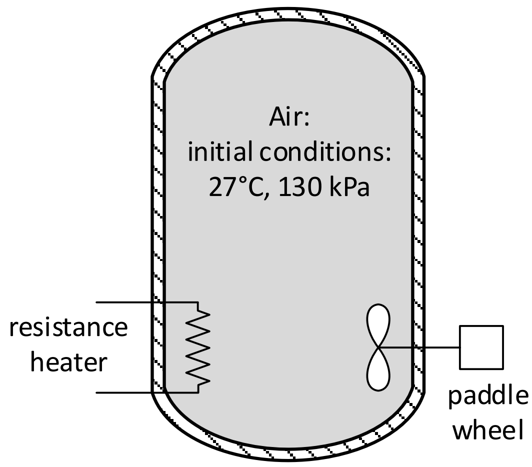


Figure 2

Question 4 (cont'd)

Question 4 (cont'd)

Question 4 (cont'd)

Useful Formulae

General & Properties

$$h = u + Pv \qquad y_x = y_\mathrm{f} + x y_\mathrm{fg} \qquad x = \frac{m_\mathrm{sat.vapour}}{m_\mathrm{sat.vapour} + m_\mathrm{sat.liquid}}$$

$$P = \rho g z \qquad v = \frac{V}{m} \qquad \dot{W}_e = VI$$

Ideal Gas

$$PV = mRT \qquad PV = NR_\mathrm{u}T \qquad R_\mathrm{u} = 8.314\,\frac{\mathrm{kJ}}{\mathrm{kmol\ K}} \qquad R = \frac{R_\mathrm{u}}{M}$$

$$\Delta u = C_\mathrm{v}\Delta T \qquad \Delta h = C_\mathrm{p}\Delta T \qquad C_\mathrm{p} = C_\mathrm{v} + R \qquad \gamma = k = \frac{C_\mathrm{p}}{C_\mathrm{v}}$$

First Law – Closed Systems

$$\underbrace{E_\mathrm{in} - E_\mathrm{out}}_{\substack{\text{net energy transfer}\\ \text{by heat and work}}} = \underbrace{\Delta E_\mathrm{system}}_{\substack{\text{change in internal,kinetic}\\ \text{potential,etc energies}}} \qquad W_{b,out} = \int_1^2 P\,dV$$

$$Q - W = \Delta U = m\Delta u \qquad Q_\mathrm{net,in} = W_\mathrm{net,out} \quad \text{(closed cycle)}$$

First Law – Open Systems

$$\dot{m} = \rho \mathcal{v} A = \rho \dot{V} = \frac{\dot{V}}{v} \qquad \mathcal{v} = \text{velocity}$$

Steady Flow Processes

$$\sum \dot{m}_\mathrm{in} = \sum \dot{m}_\mathrm{out} \qquad \dot{Q} - \dot{W} = \underbrace{\sum\nolimits_\mathrm{out} \dot{m}\left(h + \frac{\mathcal{v}^2}{2} + gz\right)}_{\text{for each outlet}} - \underbrace{\sum\nolimits_\mathrm{in} \dot{m}\left(h + \frac{\mathcal{v}^2}{2} + gz\right)}_{\text{for each inlet}}$$

$$\dot{Q} - \dot{W} = \dot{m}\left[(h_2 - h_1) + \frac{\mathcal{v}_2^2 - \mathcal{v}_1^2}{2} + g(z_2 - z_1)\right]$$

Uniform Flow Processes

$$\sum m_\mathrm{in} - \sum m_\mathrm{out} = (m_\mathrm{final} - m_\mathrm{initial})_\mathrm{system} = (m_2 - m_1)_\mathrm{system}$$

$$Q - W = \sum m_\mathrm{out} h_\mathrm{out} - \sum m_\mathrm{in} h_\mathrm{in} + (m_2 u_2 - m_1 u_1)_\mathrm{system}$$

Question 1 – (3 marks)

Under what conditions (temperature, pressure, density) could the *ideal gas equation* be used to *accurately* predict properties for *steam*?

- high temperature
- Low pressure
- Low density

3

Question 2 – (9 marks)

Fill in the missing information in the following table for ***R134a***.

	T (°C)	*p* (kPa)	*h* (kJ/kg)	*x*	Phase description
a.	-22	121.72	22.89	0	saturated liquid
b.	31.31	800	200	0.6	sat. mixture
c.	-20	132.82	89.36	0.3	sat. mixture
d.	8	1000	62.69	—	compressed liquid
e.	40	200	287.74	—	superheated
f.	85	2928.2	279.51	1.0	saturated vapour

18x ½ each = 9

b) $h = h_f + x h_{fg} \Rightarrow x = \frac{h - h_f}{h_{fg}} = \frac{200 - 95.48}{171.86} = 0.6$

c) $h = h_f + x h_{fg} = 25.47 + 0.3 \times 212.96 = 89.36 \frac{kJ}{kg}$

Question 3 – (16 marks)

A piston-cylinder device (Figure 1) with a set of stops on the top contains 2.2 kg of saturated liquid water at 150 kPa. Heat is transferred to the water, causing some of the liquid to evaporate and move the piston upwards (at constant pressure). When the piston reaches the stops, the enclosed volume is 45 l. More heat is transferred until the pressure is doubled. Determine:

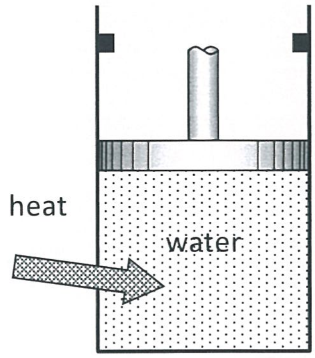


Figure 1

a) The amount of saturated liquid (if any), in kg, at the final state.

b) The final temperature.

c) The total work and heat transfer.

d) Show the process on a *p*–*v* diagram (with respect to saturation lines).

Initial conditions: (State 1)

$p_1 = 150$ kPa

$m = 2.2$ kg

$x = 0$

$u_1 = u_f = 466.97$ kJ/kg (1)

$v_1 = v_f = 0.001053$ m³/kg (1)

$\Rightarrow V_1 = m v_1 = (2.2\ \text{kg})(0.001053\ \text{m}^3/\text{kg}) = 0.0023166\ \text{m}^3$ (1)

State 2:

$V_2 = 45\ \ell = 0.045\ \text{m}^3$

$p_2 = 150$ kPa

State 3:

$V_3 = V_2 = 0.045\ \text{m}^3$

$\Rightarrow v_3 = V_3/m = 0.02045$ m³/kg (1)

$p_3 = 2 \times p_1 = 300$ kPa

$v_f < v_3 < v_g$ @ 300 kPa $\Rightarrow$ saturated mixture

$$\Rightarrow x_3 = \frac{v_3 - v_f}{v_{fg}} = \frac{0.02045 - 0.001073}{0.60582 - 0.001073} = 0.032$$ (1)

Question 3 (cont'd)

$$\Rightarrow \; u_3 = u_g + x_3 u_{fg} =$$
$$= 561.11 + 0.032 \times 1982.1 = 624.54 \ \frac{kJ}{kg} \quad (1)$$

a) $\underline{\underline{m_f}} = (1 - x_3)\, m = (1 - 0.032)(2.2\ kg) = \underline{\underline{2.13\ kg}} \quad (2)$

b) $\underline{\underline{T_3}} = T_{sat@300kPa} = \underline{\underline{133.53\ °C}} \quad (2)$

c) $\underline{\underline{W_{13}}} = W_{12} + \cancel{W_{23}}^{=0} = p_1 (V_2 - V_1) =$

$$= (150\ kPa)(0.045 - 0.0023166)\ m^3 = \underline{\underline{+6.4025\ kJ}} \quad (2)$$

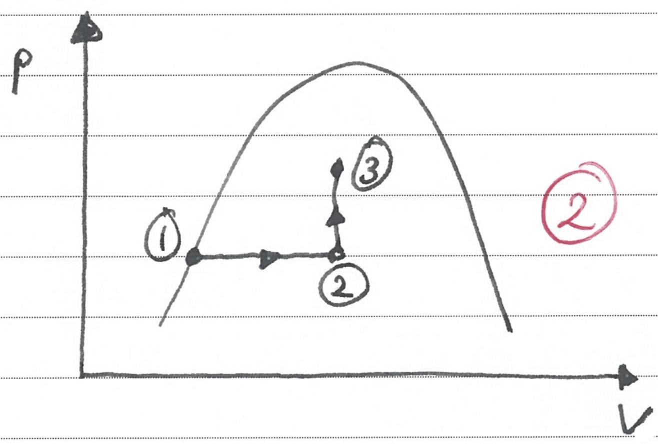
$$Q_{13} - W_{13} = U_3 - U_1 = m(u_3 - u_1)$$

$$\Rightarrow \; \underline{\underline{Q_{13}}} = W_{13} + m(u_3 - u_1) =$$

$$= 6.4025\ kJ + (2.2\ kg)(624.54 - 466.97)\ kJ/kg =$$

$$= \underline{\underline{353.1\ kJ}} \quad (2)$$

d)



(2)

Σ16

Question 4 – (8 marks)

An insulated rigid vessel with a volume of 3 m^3 is fitted with a resistance heater and a paddle wheel as shown in Figure 2. The vessel is initially filled with air at a temperature of 27°C and a pressure of 130 kPa. Now, a current of 4 A from a 230 V supply is passed through the resistance heater while the paddle wheel transfers energy at a rate of 150 J/s to the air. Determine the time taken for the air to reach a temperature of 227°C.

(State any assumptions made and also state any tables used for data).

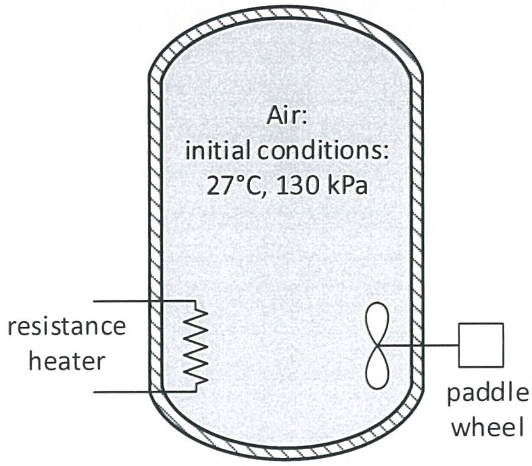


Figure 2

Initial state: $P_1 = 130$ kPa
$T_1 = 27°C = 300$ K
$V_1 = 3$ m^3

Final state: $P_2 = ?$
$T_2 = 227°C = 500$ K
$V_2 = V_1$

$$\overset{=0}{\cancel{Q}} - W = \Delta U$$

$$(-\dot{W}_{el} - \dot{W}_{sh})\Delta t = m(u_2 - u_1) \quad (2)$$

$$\dot{W}_{el} = -(4\text{ A} \times 230\text{ V}) = -920 \text{ J/s} \quad (1)$$
$$\dot{W}_{sh} = -150 \text{ J/s}$$

$$m = \frac{P_1 V_1}{R T_1} = \frac{(130 \text{ kPa})(3 \text{ m}^3)}{(0.287 \frac{\text{kJ}}{\text{kg K}})(300 \text{ K})} = 4.53 \text{ kg} \quad (1)$$

Question 4 (cont'd)

$$u_2 - u_1 = c_{v,avg}\,\Delta T =$$

$$= \left(0.726\ \tfrac{kJ}{kg\,K}\right)(500-300)\,K =$$

$$= 145.2\ \tfrac{kJ}{kg} \quad 2$$

$$T_{avg} = \frac{500+300}{2} = 400\,K$$

$$c_{v,avg} = 0.726\ \tfrac{kJ}{kg\,K}$$

(Table A-2)

or: From table A-18:

$$u_1 = 214.07\ \tfrac{kJ}{kg} \quad @\,300\,K$$

$$u_2 = 359.49\ \tfrac{kJ}{kg} \quad @\,500\,K$$

$$u_2 - u_1 = 145.42\ \tfrac{kJ}{kg}$$

$$\Rightarrow \quad \Delta t = \frac{m(u_2-u_1)}{\left(-\dot{W}_{el} - W_{sh}\right)} = \frac{(4.53\,kg)\left(145.2\ \tfrac{kJ}{kg}\right)}{(920+150)\times 10^{-3}\ \tfrac{kJ}{s}} =$$

$$= \underline{614.7\,s} = 10\text{ minutes } 14\text{ seconds} \quad 2$$

$\Sigma\, 8$